

Attribute value	Temperature		Display as	
-545	-5.45	22.19	-5.5	22
-1823	-18.23	-0.81	-18	-1

Sample Conversion Code

Sample code provided to ensure consistent Fahrenheit to Celsius and vice-versa conversion between devices and across vendors.

For degF: the value is a int8u representing 2x temperature value in Fahrenheit (to get 0.5 resolution).

For degC: the value is a int16s representing Celsius in 0.01 resolution as expected by the Matter format.

```

/*
 * Function : translateMatterTempDataType()
 * Description : Converts the temperature setpoints in Matter
 * to the half degF format.
 * The half degF format is 8-bit unsigned,
 * and represents 2x temperature value in
 * Fahrenheit (to get 0.5 resolution).
 * The format used in Matter is 16-bit signed
 * in Celsius and multiplied by 100
 * to get 0.01 resolution.
 * e.g. 2500 (25.00 deg C) ---> 0x9A (77 deg F)
 * Input Para : Temperature in Matter (degC) format
 * Output Para: Temperature in half DegF format
 */
int8u translateMatterTempDataType(int16s temperature)
{
    int32s x = temperature;
    // rearrangement of
    // = (x * (9/5) / 100 + 32) * 2;
    // the added 250 is for proper rounding.
    // a rounding technique that only works
    // with positive numbers

    return (int8u) ((x*9*2 + 250)/ (5*100) + 64);
}

/*
 * Function : translateDegFTemp

```